

AERODYNAMIC ASPECTS OF A HIGH ALTITUDE LONG ENDURANCE UAV DEVELOPMENT

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ABSTRACT

The present paper summarizes the preliminary design cycle and main technical milestones in the process of development of a high altitude, long endurance (HALE) UAV of large dimensions, in the range of 7500 kg maximum takeoff weight. The main aspects dealt with, are the aerodynamic ones and the integration of the engine in the airframe.

The problematic of flying at high altitude at a high subsonic regime and the solutions adopted are exposed.

The theoretical and experimental activities performed in order to define, design and validate the viability of the configuration are briefly described.

The air vehicle presented has several unique features, such as laminar high aspect ratio wing, inlets integrated into the wings close to the leading edge, etc.

This paper is a synthesis of the feasibility work done and stresses the main highlights.

By the end of the preliminary development cycle the results of the experimental work showed that the UAV configuration, as defined, has sound aerodynamic characteristics, meets the overall performance requirements and may be a solid basis for further development.

INTRODUCTION

The potential and the versatility of use for a large dimensions UAV flying at high altitude for a long period of time, became

lately well recognized by the potential users of such a vehicle and also by the aeronautical industries capable of developing such a complex aircraft.

When considering the mission requirements for a HALE UAV it is evident that several critical technologies, in a stage of sufficient maturity, are needed in order to be able to accomplish such a demanding task.

In order to be able to fly at an altitude of about 60000 ft for a long period of time and to carry a significant payload, a highly efficient aerodynamic configuration, in terms of lift over drag ratio, with a wing of high aspect ratio, has to be designed. (Several such configurations are already successfully flying: Global Hawk, Voyager, etc.). A delicate balance between the vehicle drag, weight and engines capabilities (thrust and specific fuel consumption) must be kept. The structural design has to be such that, while keeping the weight to a reasonable minimum, it has to comply with numerous, and sometime contradictory, stringent requirements: rigidity, aero-elasticity and flutter, structural and system installation, etc.

An appropriate engine, adapted to high altitude operation and properly integrated in the airframe has to be chosen.

IAI is well known for its capability of developing and producing families of reliable UAV, with a constant trend of increase in weight, altitude and payloads capabilities.

Fig. 1 shows a chart of this family, in which the constant trend of increased capabilities is evident. The HALE characteristics are also inserted, standing well above all the other UAV.

DESIGN CONSIDERATIONS

Several configurations were considered during the preliminary design study. The feasibility study and the down selection process between the different configurations investigated in this study are beyond the scope of this paper.

The sizing process of the UAV took into account that in order to fly at high altitude, about 60000 ft, at a low subsonic Mach number of about $0.55 \div 0.60$, several prevailing parameters have to be considered:

- a. The ceiling factor, $M^2 \cdot CL_{max}$ has to be as high as possible (and therefore the maximum usable lift has to be as high as possible), for obtaining maximum flight altitude. For the configuration selected a lift coefficient of about 1.0 is needed at loiter.
- b. In order to obtain maximum performance, a high value of the lift over drag ratio is needed (L/D_{max}). A goal of 33 was decided upon for the selected configuration in order to meet the mission requirements.

The resulting conclusion, from the above mentioned considerations, was that in order to meet the performance requirements a laminar wing has to be designed and incorporated in the configuration. This wing has to be designed for high lift at the low Reynolds number regimes prevailing at high altitude. Fig. 2 and 3 show charts illustrating the achievable domain for these parameters.

HALE UAV CONFIGURATION

The selected configuration (resulting from the feasibility study and down selection process) on which most of the theoretical and experimental work was performed is schematically shown in Fig. 4.

The HALE UAV configuration consists mainly of four elements:

- a. Inner wing - A 35° swept wing, accommodating the major systems: two

turbofan engines, the fuel tanks, landing gears, etc.

- b. Outer wing - A 5° swept wing (practically a two dimensional wing), with four ailerons, two per side.
- c. Tail boom - Supporting the configuration tail and including an additional fuel tank.
- d. V tail - Which stabilize the configuration and featured two monoblock deflectable surfaces, used for longitudinal and directional control.

The convergence toward this configuration was done during several design cycles.

The engine incorporated in this design is the Williams FJ44 -2E type (SE3 derivative, adapted for high altitude operation), with a static thrust at sea level of 2300- 2600 lb.

HALE UAV ANALYTIC DESIGN

Wing Design

The HALE UAV configuration is basically a "flying wing with a tail installed on the boom". The wing consists of two parts : the outer, two dimensional wing, and the inner three dimensional, swept wing. As said, in order to meet the performance requirements the approach was to design both the outer and the inner wing as laminar wings.

For the outer wing an airfoil was designed with the following main requirements:

- High thickness ratio for structural stiffness
- Natural laminar flow at the design point, namely, low transonic Mach number and low Reynolds number
- High maximum lift coefficient in order to maximize the ceiling parameter and the maximum lift to drag ratio.

Fig. 5 presents the aerodynamic characteristics of the outer wing airfoil.

In order to enhance the longitudinal stability of the configuration, the outer wing was swept backward by 5° .

The concept of natural laminarization of the inner wing is more complex and is beyond the scope of this paper; only brief high lights will be given here.

For the inner wing the target was to develop a wing with turbulent characteristics at low altitude and to ensure the transition of the flow regime to laminar characteristics at an altitude close to the loiter altitude (the Mach number is increasing and the Reynolds number is decreasing with altitude increase). The main issue to be dealt with, is the boundary layer stability. Airfoils with the general rules for preventing premature amplification of instability and adapted to the requirements of the UAV were designed, and incorporated in the inner wing.

Inlet Design

Another major activity in the definition of the HALE UAV configuration was the design, analysis and test of the inlet, i.e. the engine/ airframe compatibility.

As mentioned, the engine installed in the airframe was the Williams FJ44-2E turbofan engine. The inlet was designed according to the engine specifications and requirements, and matched for two design points: takeoff and end of climb (maximum engine power). Several off design points were also taken in consideration.

The main problematic with this kind of inlet is its proximity to the leading edge of the inner wing, resulting in a three dimensional flow at the inlet entrance. This strong three dimensional flow is prone to separation inside the short and curved duct. The engine inlet (lip and duct) was designed during several iterations by using MGAERO code[1-3]. This code is a non-viscous steady Euler equations solver. It was anticipated that by designing the inlet with reduced local pressure gradients in the

region of the entrance, the problem could be attacked by steady flow analysis.

The criterion for design was to obtain a subsonic smooth flow, without any apparent separation phenomena, inside the duct and around the inlet lip, for the two defined design points. The basic design parameters were a Mach number of 0.7 at the throat section and a ratio of 1.3 between the capture and the throat area.

The design cycle included four geometric iterations.

Fig 6 shows the flow pattern at the inlet entrance, at the beginning of the design cycle (iteration #1) and at the end of the design cycle (iteration #4) for Mach = 0.55, angle of attack of 2 deg. By redefining the lip and duct geometry, the detached flow (reversed flow) observed with the initial geometry was corrected to attached and smooth flow on the final geometry.

EXPERIMENTAL VALIDATION

In order to validate the theoretic evaluations the following wind tunnel tests were conducted :

- a. Outer wing airfoil test.
- b. Configuration test at low speed.
- c. Configuration test at high speed.
- d. Inlet test.

Airfoil Wind Tunnel Test

The designed airfoil (mainly by MSES analytical code) was tested at Ohio State University (OSU) wind tunnel on an instrumented two dimensional model with pressure taps. The airfoil was tested at conditions close to the real flight conditions. The experimental results validated the analytical design.

Fig. 7 presents comparisons between the theoretic results and the results obtained at OSU wind tunnel.

Low Speed Wind Tunnel Test

The purpose of the low speed test was to obtain the general longitudinal and lateral

characteristics of the configuration, including the tail control effectiveness.

The low speed wind tunnel test was performed on a 1:11 model in the low speed wind tunnel at IAI (force and moments).

Fig. 8 shows the model installed in the wind tunnel.

Fig. 9 presents the lift and pitching moment behavior of the basic configuration.

Fig. 10 presents the lateral / directional stability of the basic configuration.

The data obtained from the low speed test was used in order to build the data base of the configuration and to calculate its basic characteristics in flight.

This test showed that the UAV configuration was a sound one and did not have any major inherent deficiencies.

High Speed Wind Tunnel Test

The purpose of the high speed test was to validate the drag characteristics (or in other words the L/D max characteristics) at the main design point, i.e. at loitering at high altitude. In addition the flow pattern over the inner and outer wing parts was checked, to determine whether the laminar design goal was achieved.

The high speed test (force and moments) was conducted at the NLR high speed tunnel, which has good flow qualities in terms of turbulence at high speeds, provides sophisticated visualization techniques for laminar flow checks, and, by changing the pressure conditions, makes possible to vary the test Reynolds number.

The main test points were at:

Mach = 0.45; Reynolds = 14.6 million

Mach = 0.50; Reynolds = 12.0 million

Mach = 0.62; Reynolds = 9.6 million

Because of wind tunnel test chamber limitations, the outer wing model was truncated (scale model 1:10).

Fig. 11 shows the model installed in the tunnel.

An analytical procedure, based on MGAERO and MSES [4-5] codes, was

elaborated in order to transfer the test results (for the truncated model) to the full span model.

This procedure is based on two facts :

- a. Outer wing - is predominantly two dimensional.
- b. Inner wing - there is a well defined region on the truncated model with aerodynamic characteristics (lift and pressure distribution) identical to those of the full span model.

Since this procedure correlates between truncated and full span theoretic models by using theoretic codes, the validity of the theoretic evaluations was demonstrated by comparing the wind tunnel test results to the truncated wing theoretic results.

Fig. 12 presents a comparison between predicted and measured pressure distribution, lift and drag for the truncated wing model.

Fig. 13 shows the L/D parameter, at Mach = 0.62, for the full aircraft (test data extrapolated to the full configuration), for the free and forced transition case. A L/D max of about 37.5, at $CL = 1.0$, is obtained in the test for the free transition case.

We have to keep in mind that the results obtained in the test are for an "ideal clean" aircraft. When additional effects are taken into account (scoops, protuberances, etc) a L/D max curve as shown in Fig. 14 has been obtained. It is seen that the goal of $L/D \text{ max} = 33$ has been practically achieved.

During the test series three visualization techniques were used:

- a. Infrared technique, based on the recording of a slight difference in the adiabatic wall temperature between laminar and turbulent flow. If the flow and the model are at different temperatures, heat from and to the model transfers quicker in the turbulent flow. This phenomenon can be observed only if the model surface has a low thermal conductivity, which can be obtained by applying an insulating

layer on the model surface. The difference in temperatures between the laminar and turbulent boundary layer is visualized by using a remote control infrared scanner.

- b. Sublimation technique, based on the property that solid crystals of certain substances (in this case acenaphtene) sublimates quicker in turbulent flow than in laminar flow.
- c. Oil technique, based on oil flow pattern on the model surface.

The infrared visualization (fig. 15) showed that for $Mach = 0.62$, $Reynolds = 9.6$ million, the flow regime for the outer wing is laminar, and for the inner wing is laminar on the upper surface and turbulent on the lower one.

Inlet Test

The last inlet iteration, obtained in the analytical study, was tested in a high speed test in the high speed wind tunnel at IAI. The main purpose of the test was to obtain the pressure recovery value at the engine face and to check the quality of the flow along the duct and on the lips.

For this purpose a half model was designed and manufactured, instrumented with a pressure rake at the engine face plane and with static pressure taps on the lips and along the inlet duct. Fig. 16 presents the inlet model.

The test was performed for a complete matrix of parameters (mass flow ratios, Mach numbers, Reynolds numbers, angle of attack).

Fig. 17 shows the pressure recovery parameter for a typical working point. At the relevant mass flow ratio of the engine a pressure recovery value of 98.5% is obtained, which is very satisfactory for this kind of inlet and does not degrade the installed characteristics of the engine.

SUMMARY

This paper presents the development cycle of a High Altitude Long Endurance UAV, covering the aerodynamic aspects. The paper is a synthesis of the theoretical and experimental activities.

By the end of those activities it can be concluded that the configuration, as defined, is a sound one and is able to perform the required mission.

This platform, or a similar one, can be used for a large diversity of missions.

ACKNOWLEDGMENT

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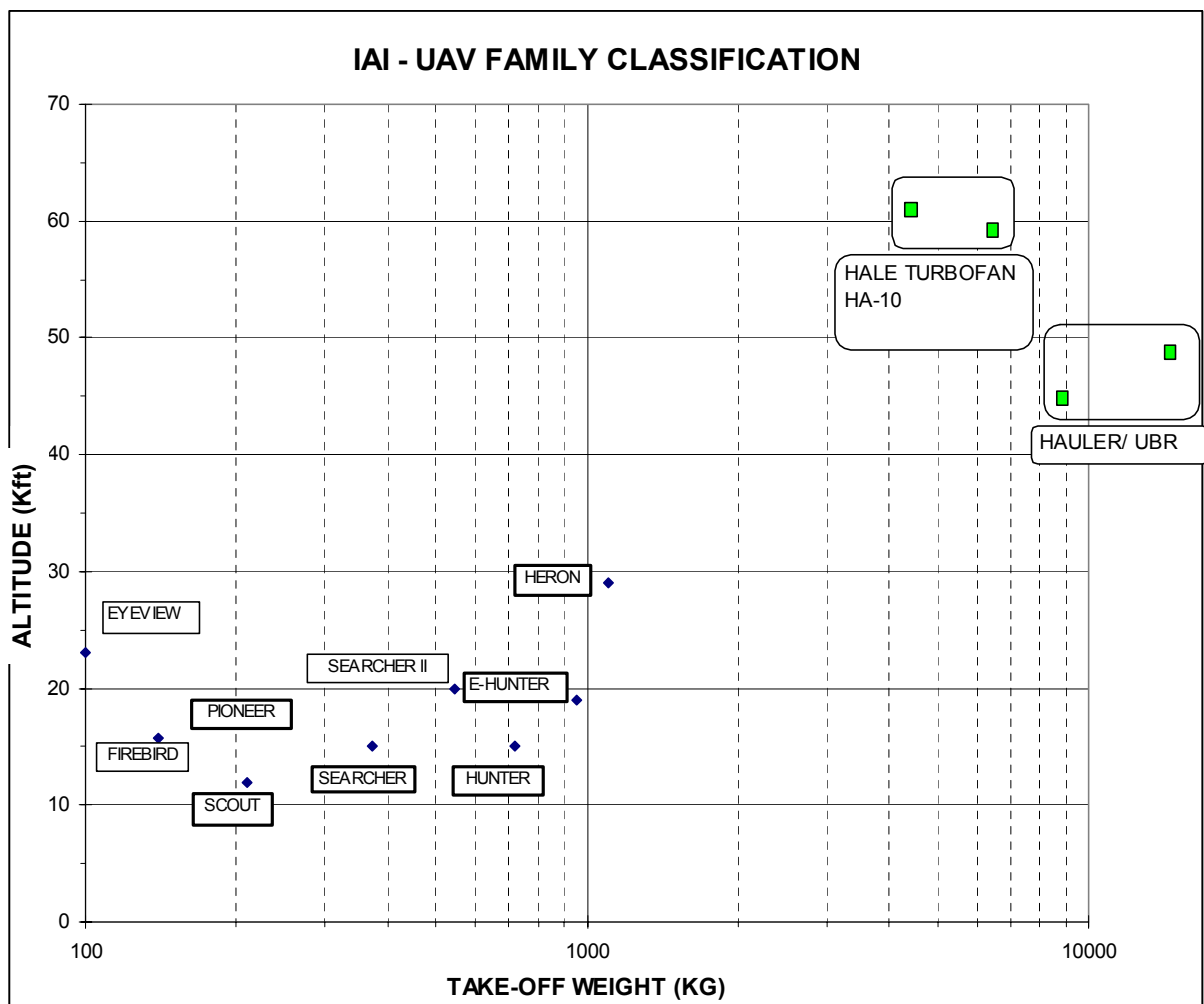


Fig 1 : UAV Family Developed at IAI

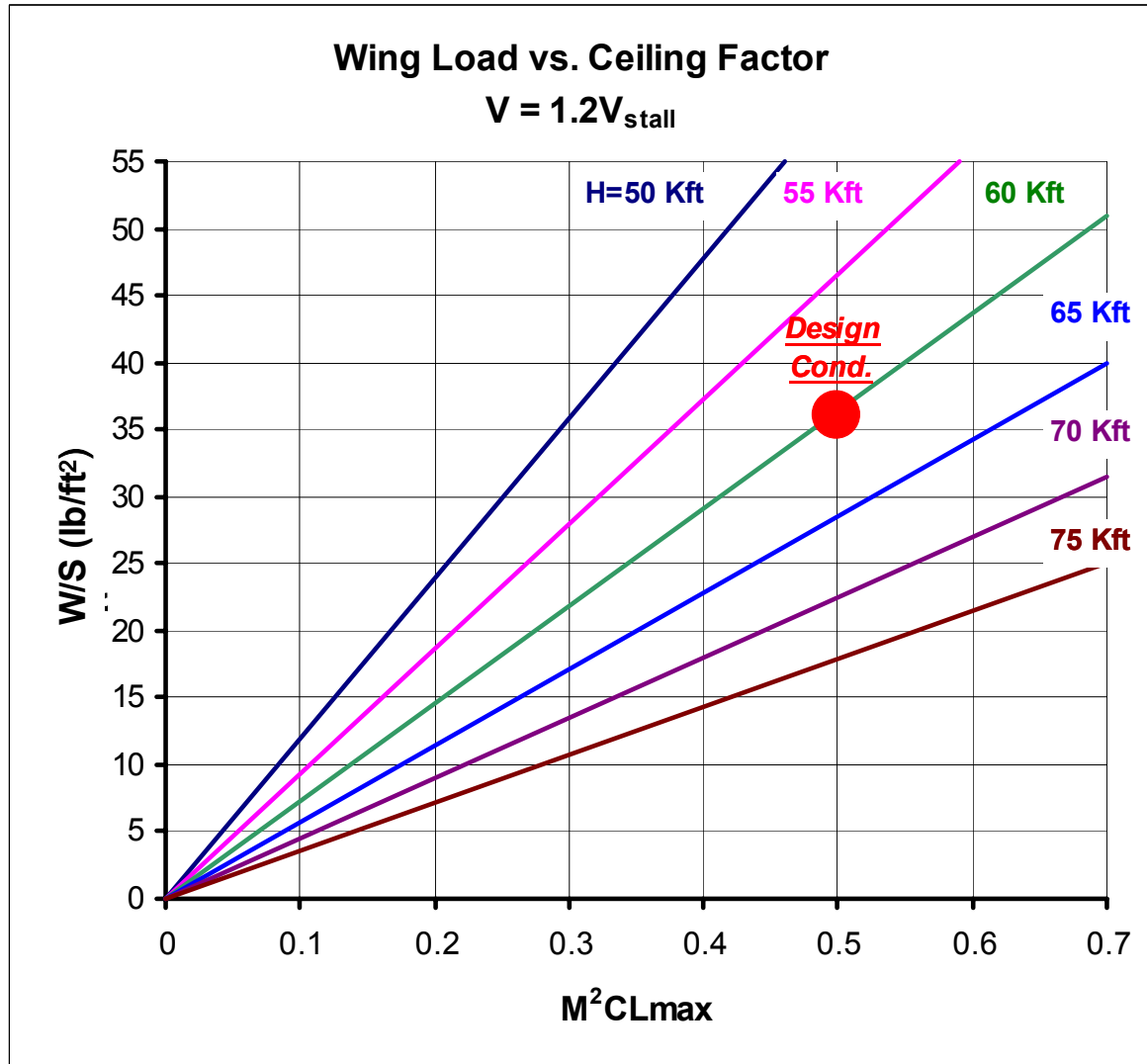


Fig. 2 : Wing Load vs. Ceiling Factor

AERODYNAMIC EFFICIENCY vs WING SPAN PARAMETER

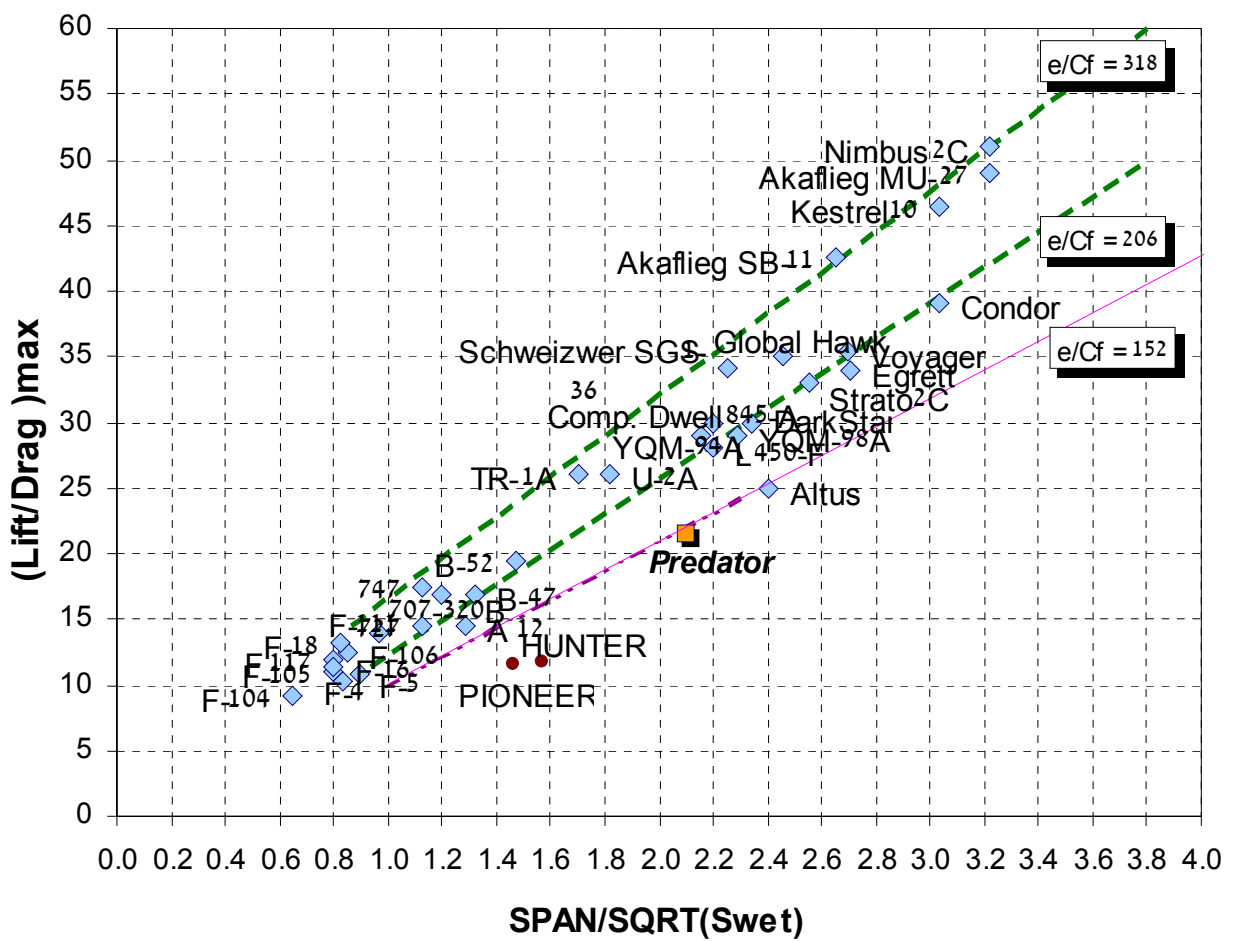


Fig. 3 : Lift over Drag Ratio vs. Wing Load

Wing span	103 ft
Gross wt.	16000 lb
Oper. Ceiling	65000 ft
Power plant	WI FJ44-2A-SE3
Static thrust	2337 (F.R.) lb

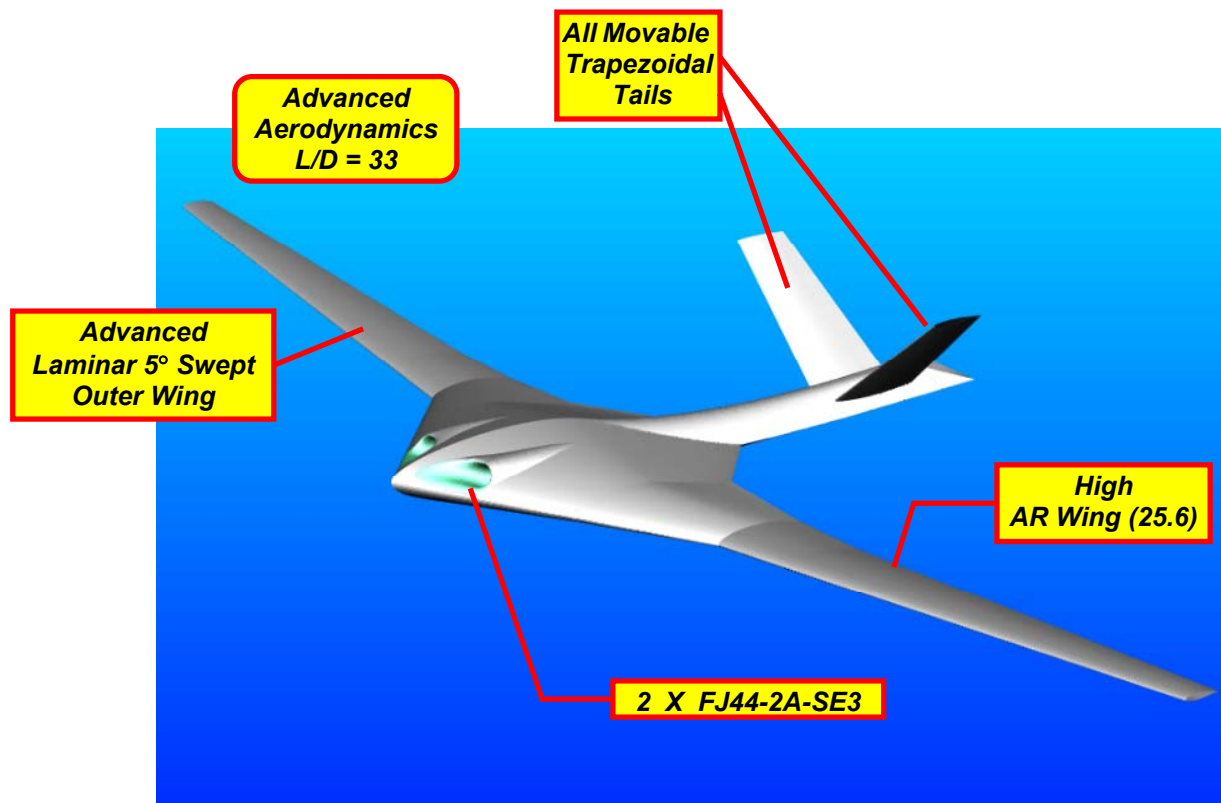


Fig. 4 : HALE UAV General View

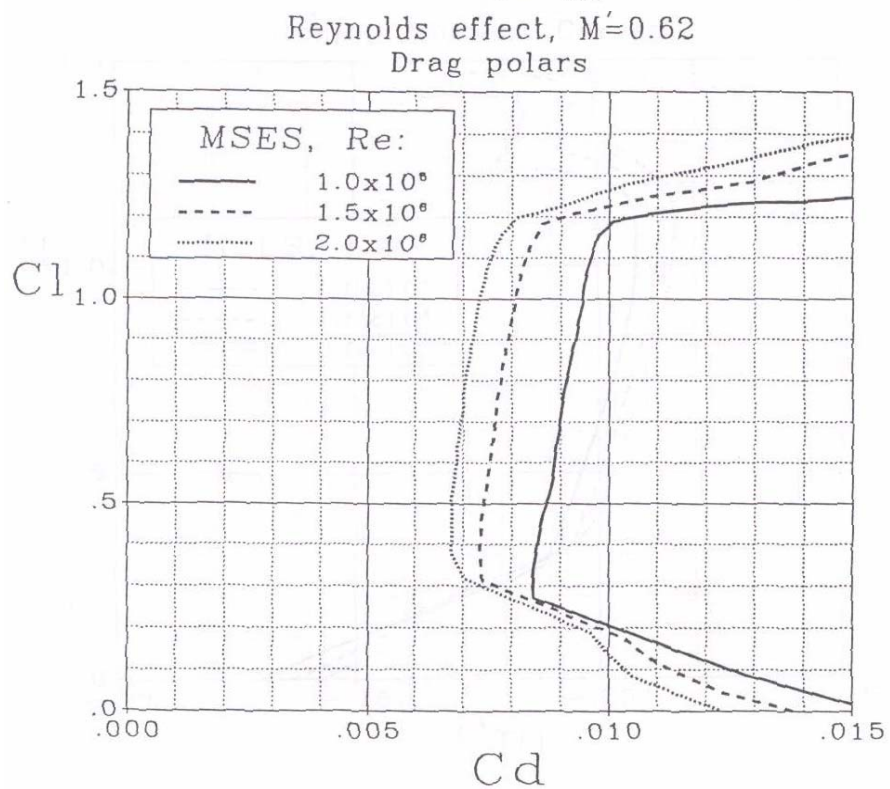
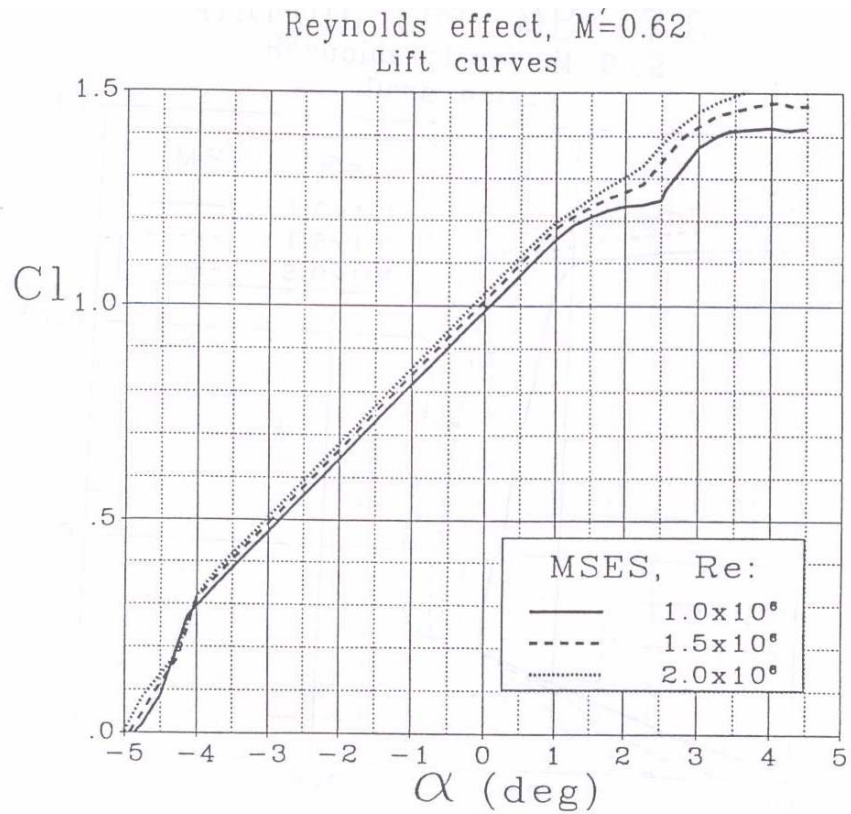
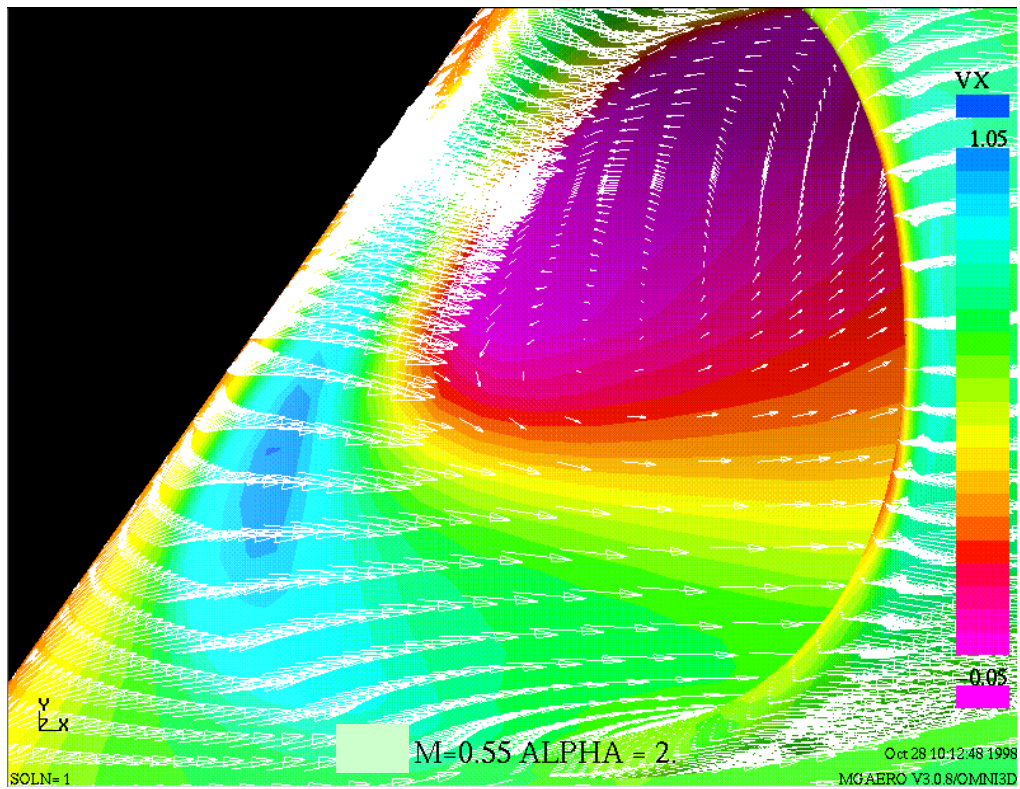
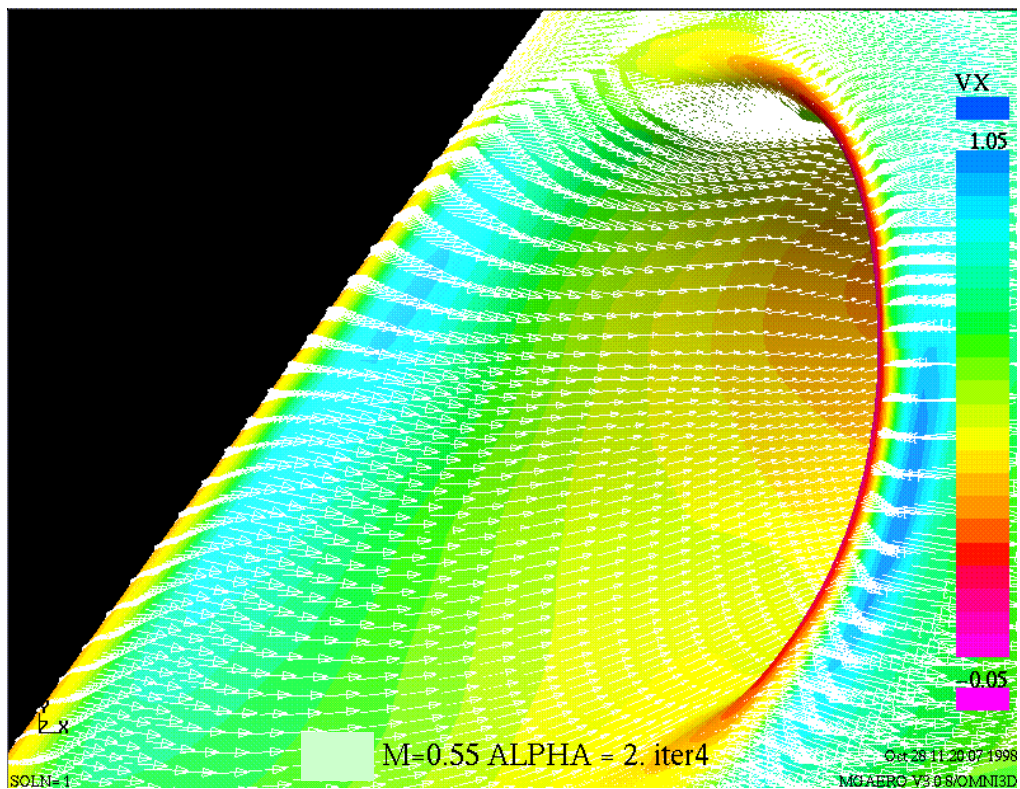


Fig. 5 : Outer Wing Airfoil Characteristics



First Iteration



Final Iteration (#4)

Fig. 6 : Inlet Design - Axial Velocity Component and Flow Pattern on Inlet Surface

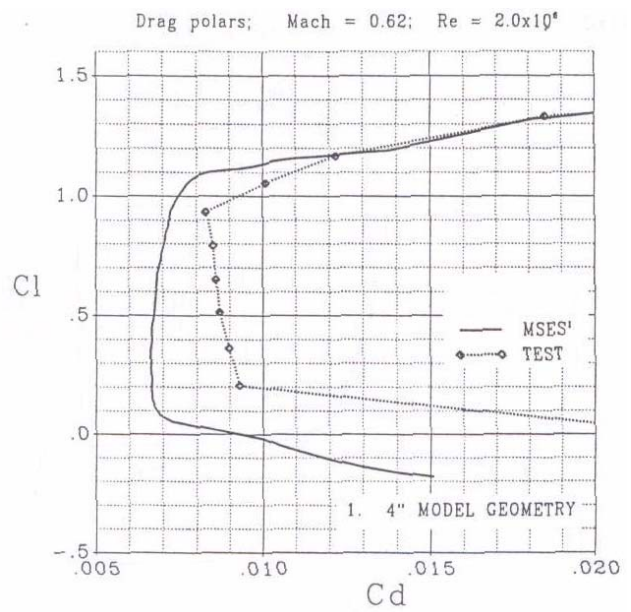
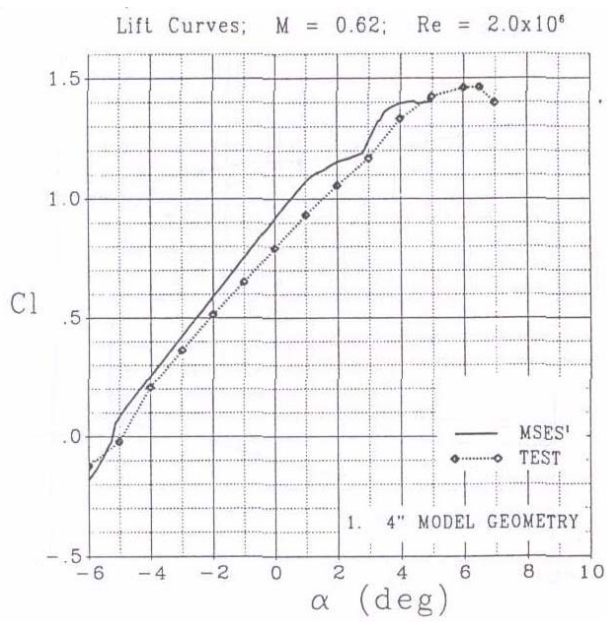
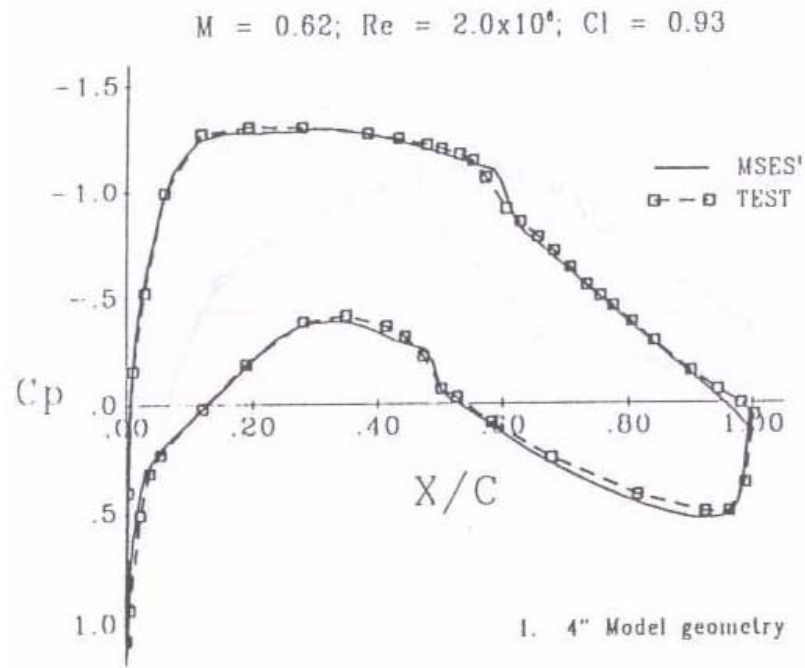


Fig. 7 : OSU Results



Fig. 8 : General View of Model Installed in the Low Speed Wind Tunnel (IAI)

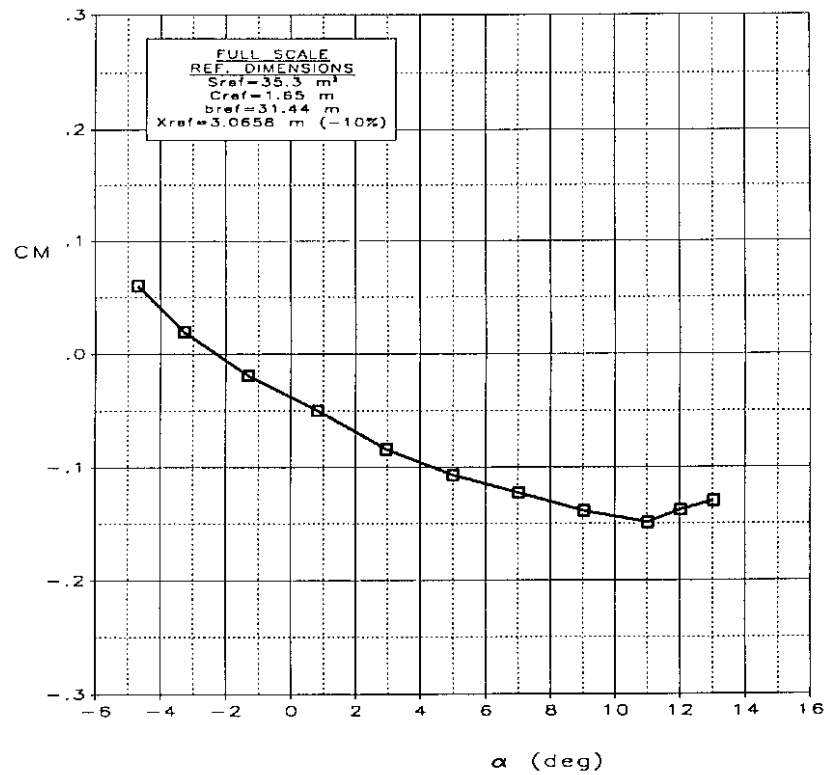
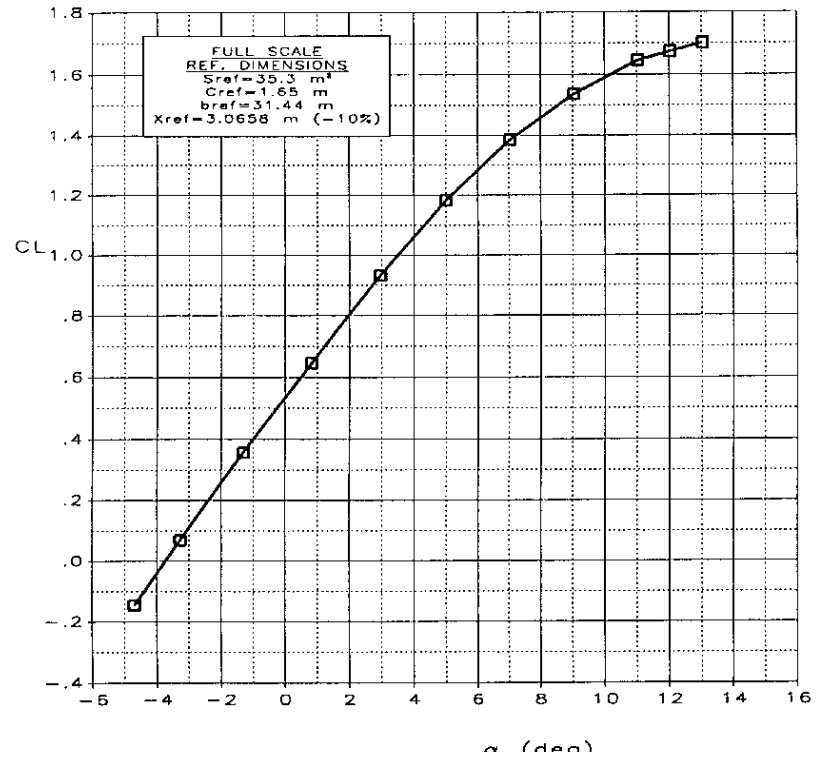


Fig. 9 : Longitudinal Characteristics

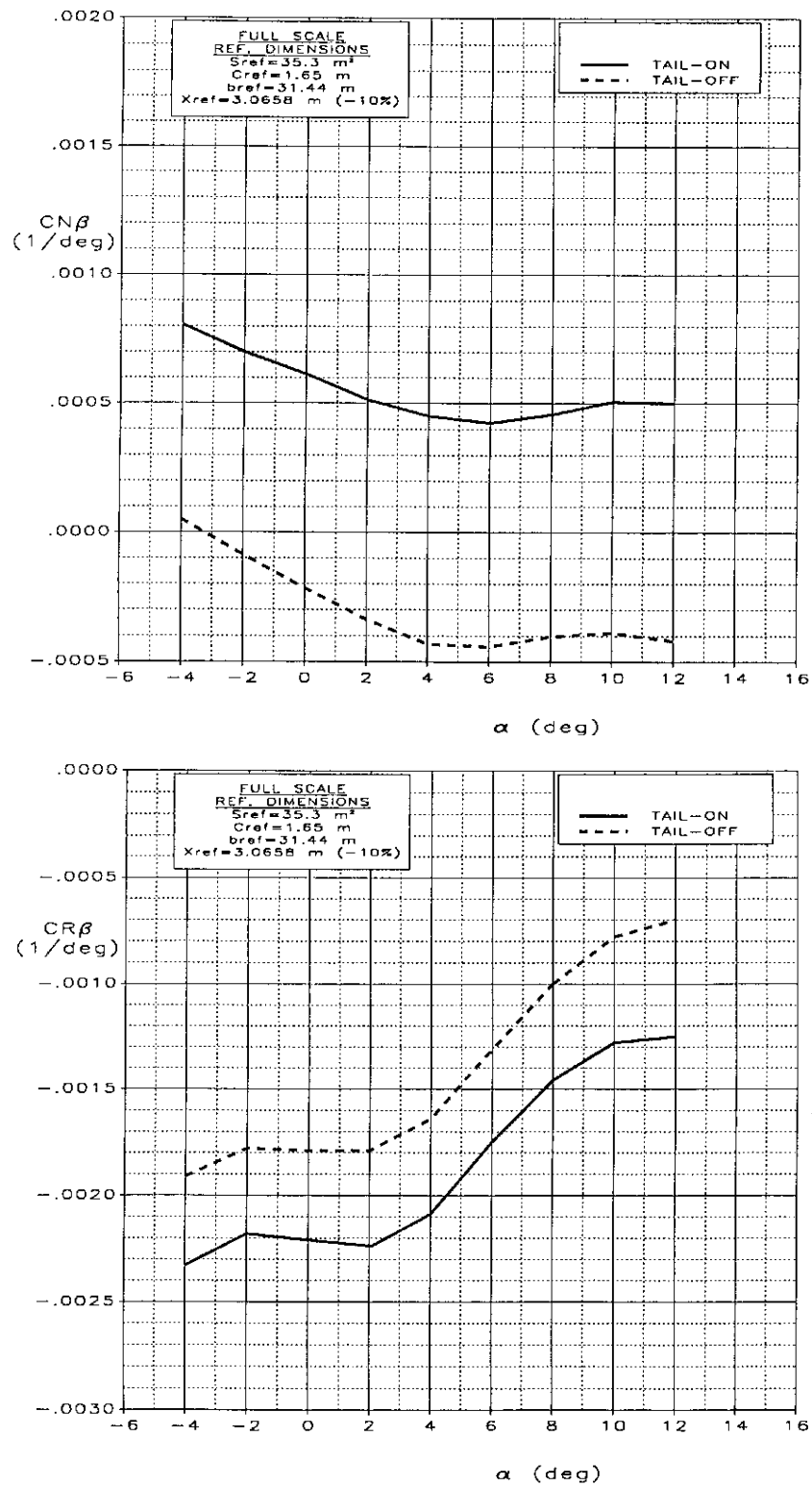


Fig. 10 : Lateral / Directional Stability

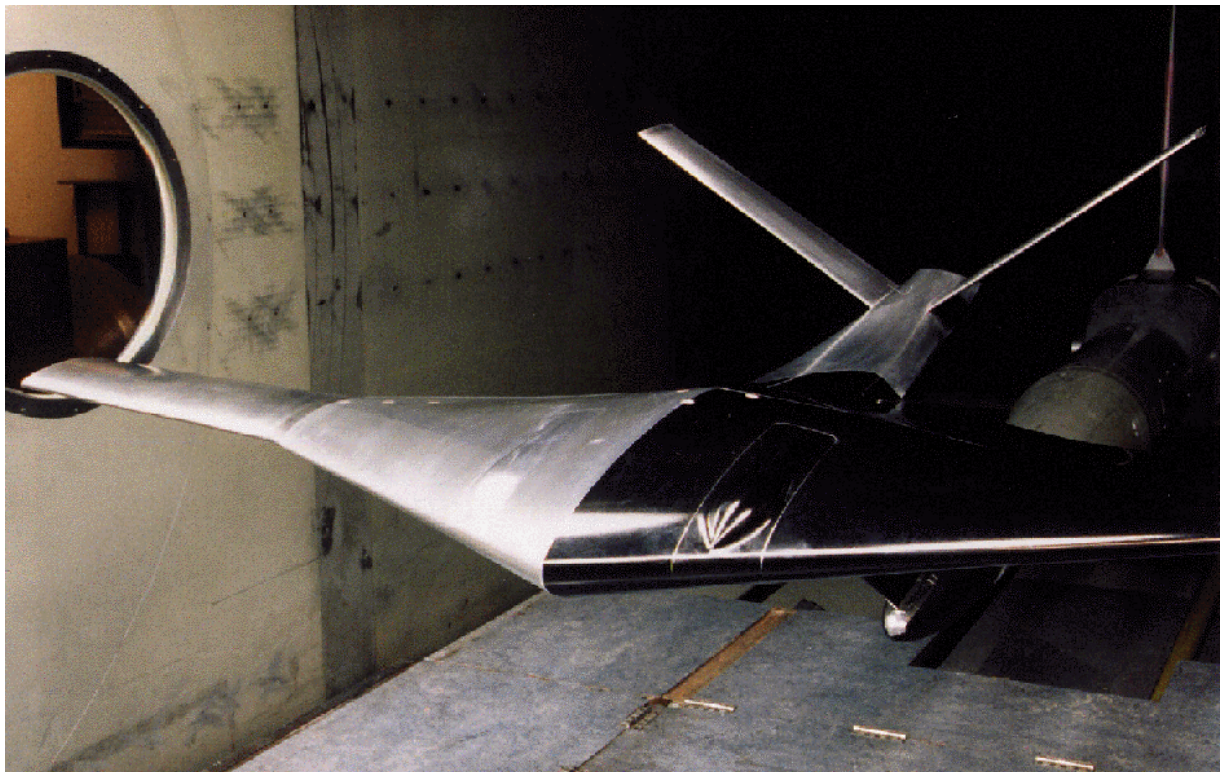


Fig. 11 : High Speed Model Installed in Wind Tunnel

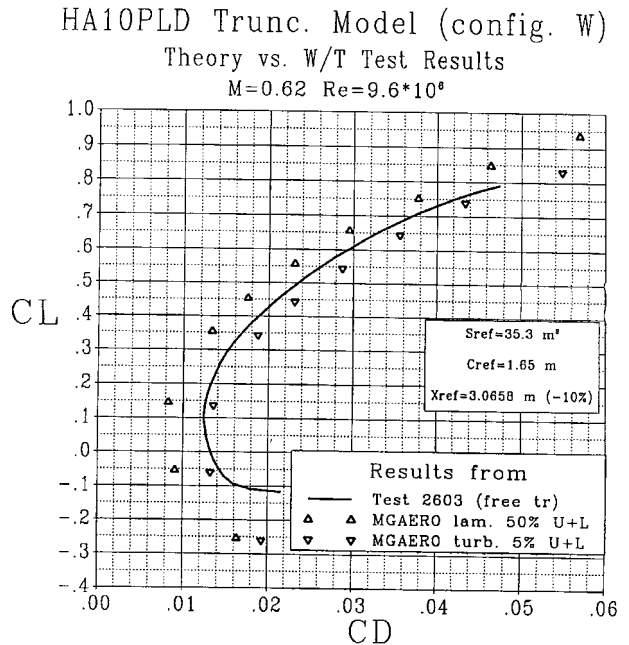
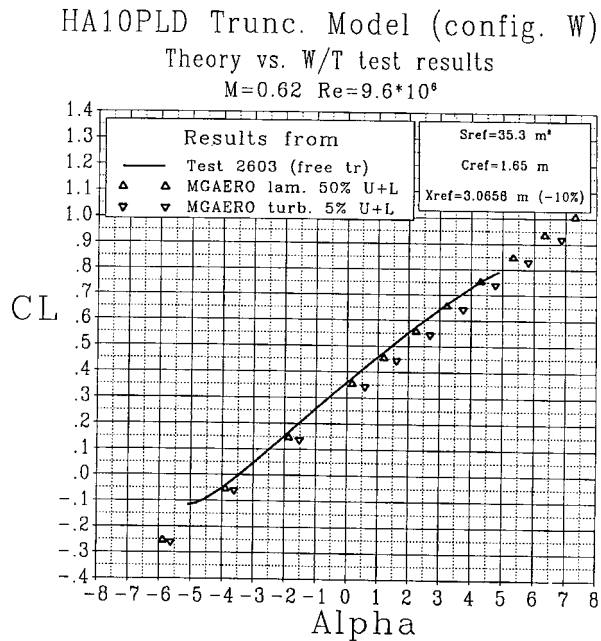
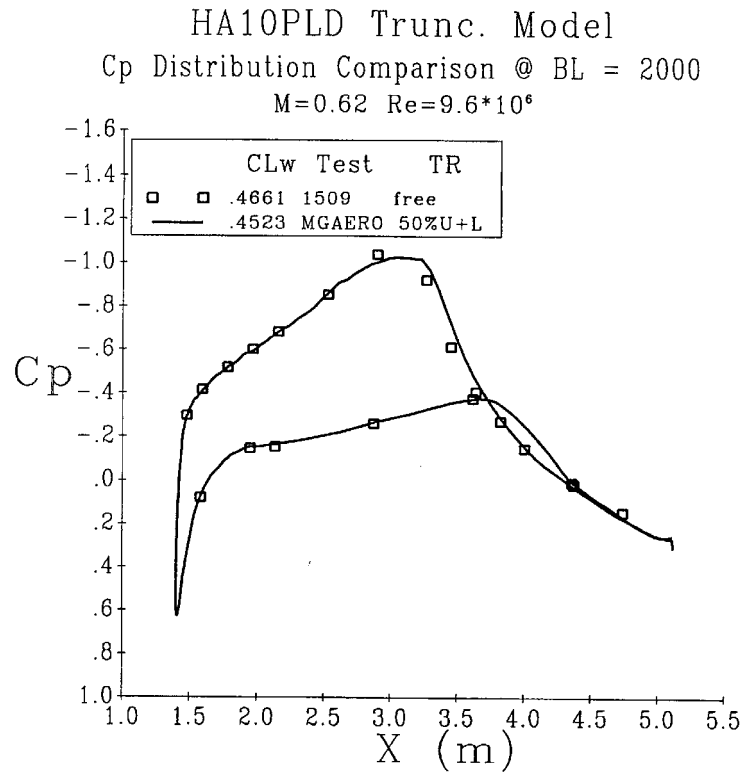


Fig. 12 : Theory vs. W/T test results at M = 0.62 Re = 9.6 * 10⁶

HA10PLD Full Model (config. W,B,T)

Transition Effect

$M=0.62$ $Re=9.6 \cdot 10^6$

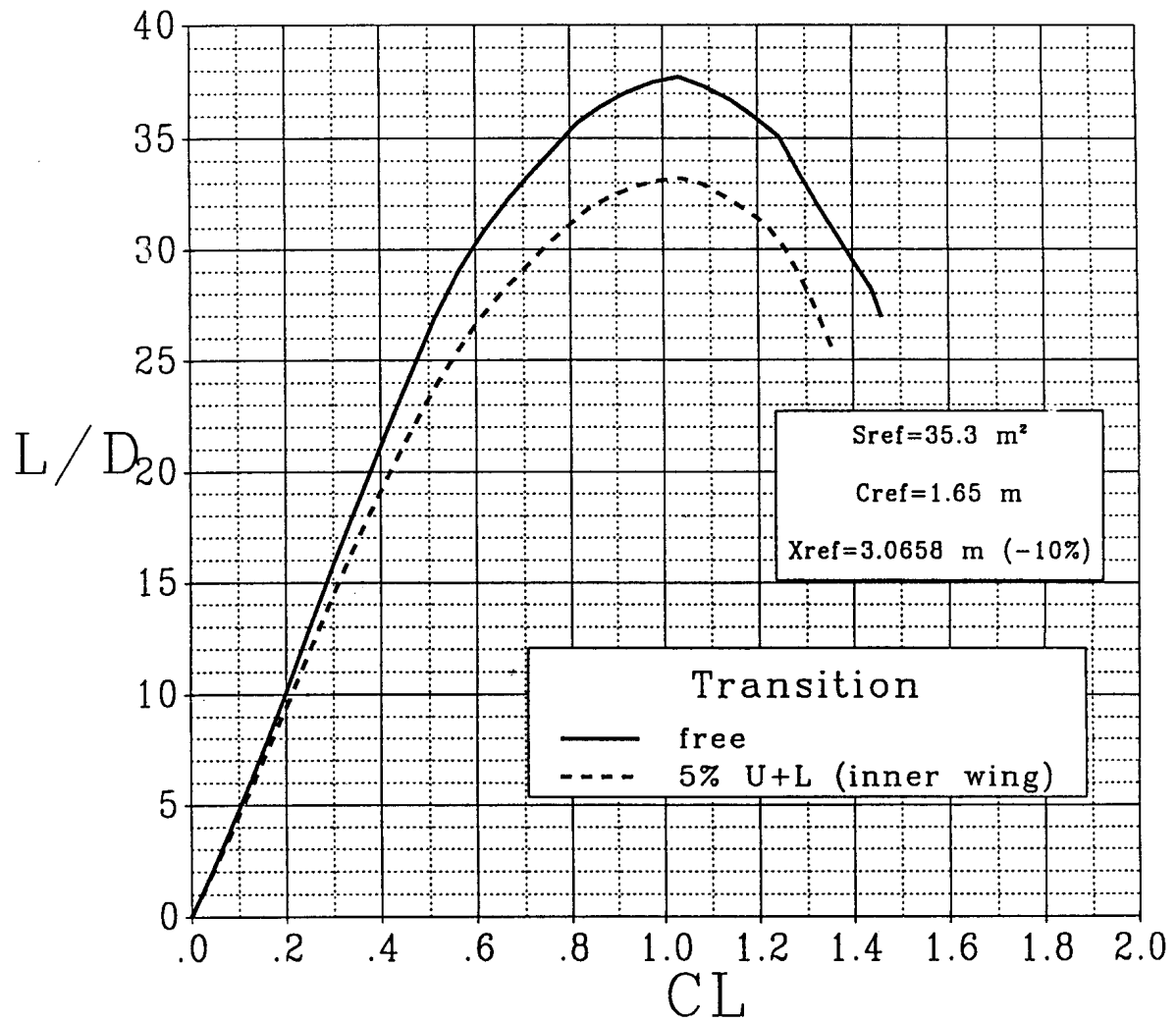


Fig. 13 : L/D parameter for "full" Wind Tunnel Model

HA10PLD

Data Base for Performance

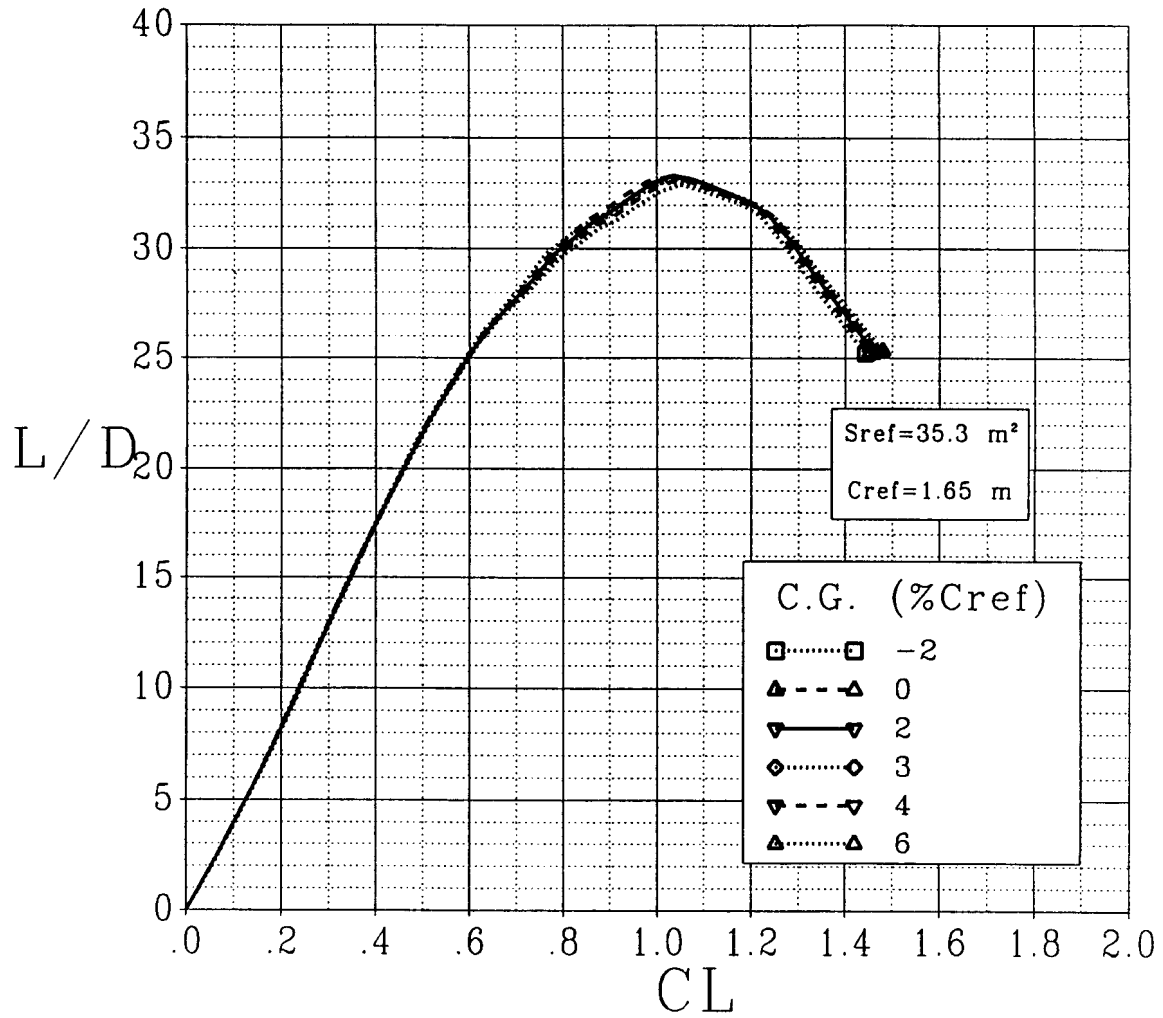
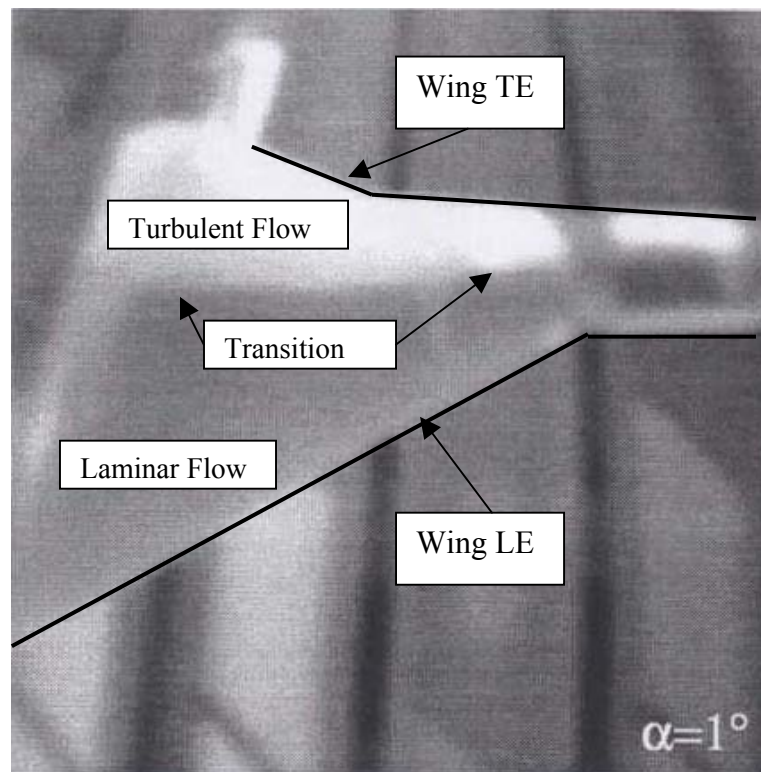
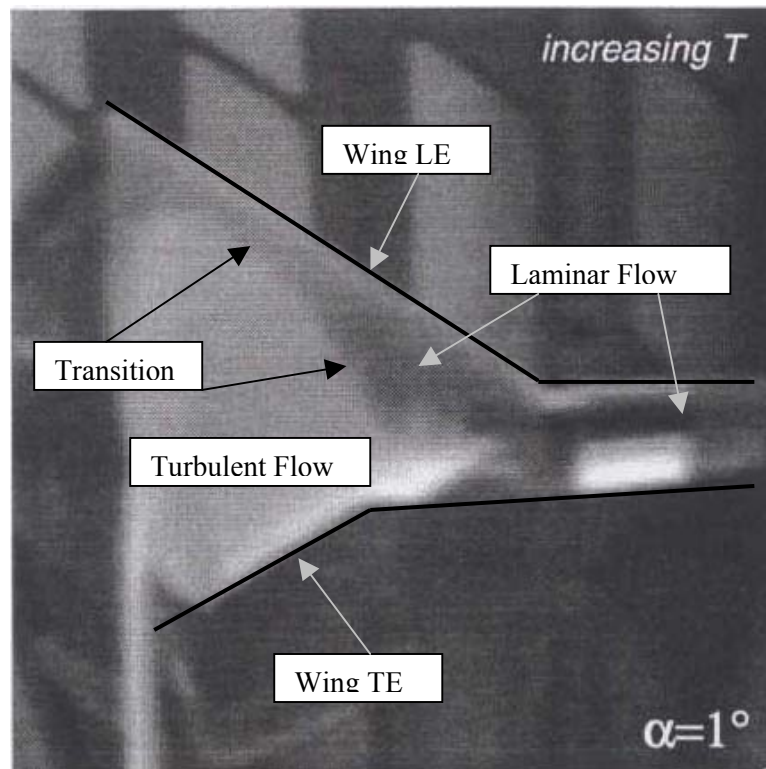


Fig. 14 : L/D Parameter for HALE UAV



Upper Surface



Lower Surface

Fig. 15 : IR visualization at $M=0.62$ $Re = 9.6 \times 10^6$ $\alpha=1^\circ$

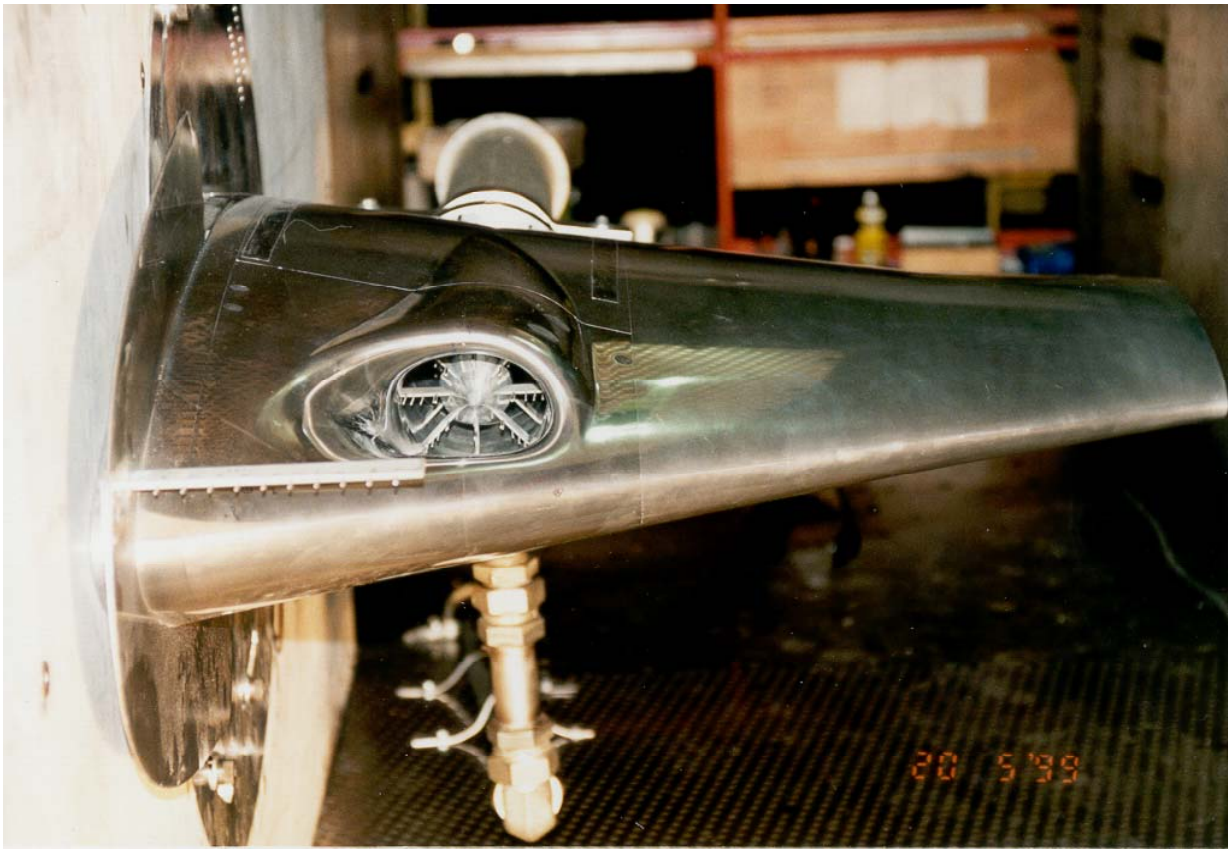


Fig. 16 : Inlet Wind Tunnel Model

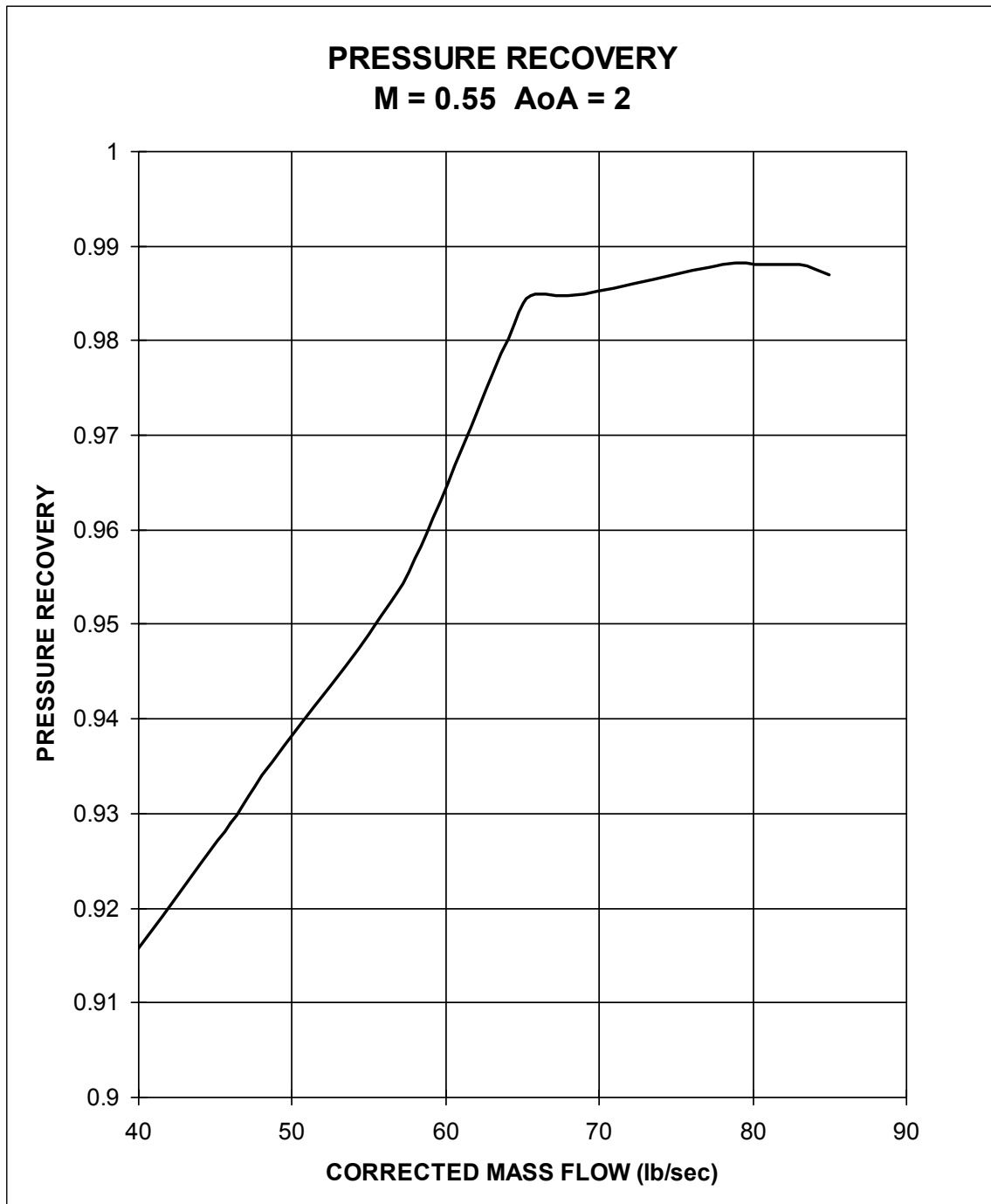


Fig. 17 : Inlet Pressure Recovery, Test Results